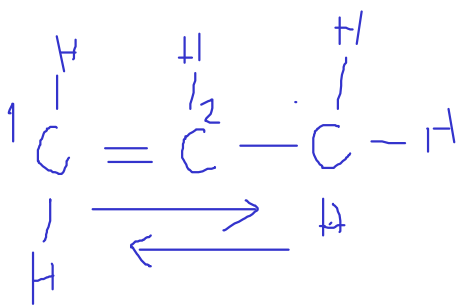
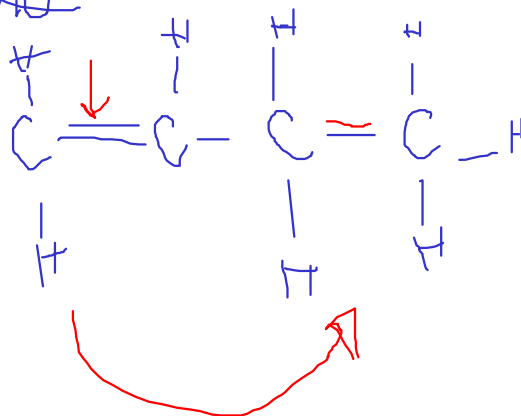


propene
3



but-1-ene
but-3-ene



Grupos orgânicos substituintes

Quando espécies orgânicas aparecem substituindo um ou mais átomos de hidrogênio de uma cadeia de hidrocarbonetos, elas recebem o nome de Grupos orgânicos substituintes. São as ramificações das cadeias carbônicas.

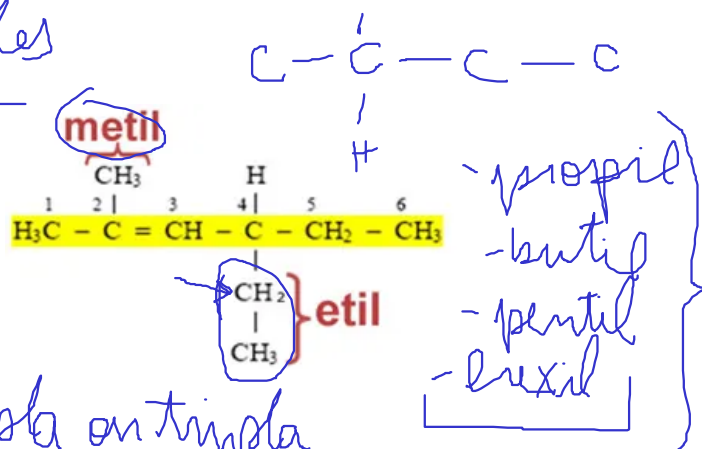
Grupos saturados

ligação simples

Prefixo que indica a quantidade de carbonos



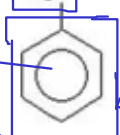
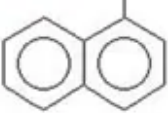
il ou ila



Grupos insaturados

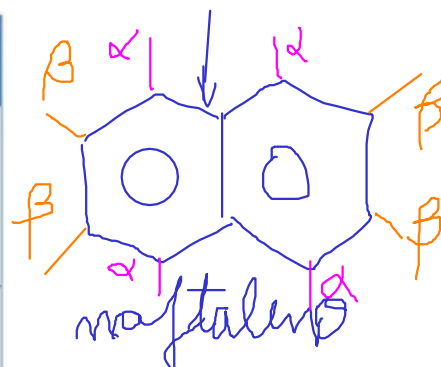
ligação dupla ou tripla

Alguns grupos orgânicos substituintes derivados de compostos insaturados e de compostos aromáticos

$\text{H}_2\text{C}-$ 	Benzil
	α -naftil
	β -naftil
	Fenil
$\text{H}_2\text{C}=\text{CH}-$	Vinil

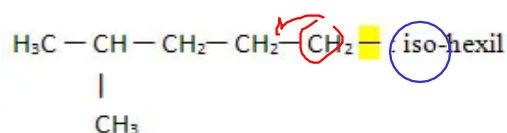
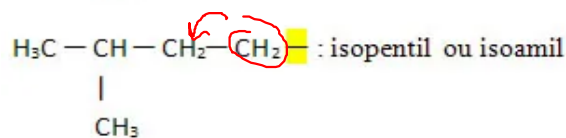
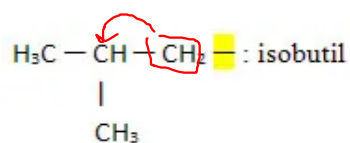
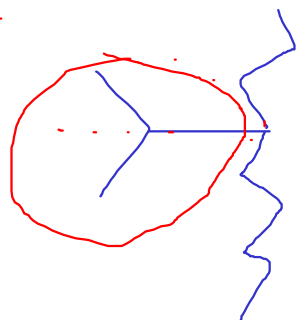
aromática

benzeno

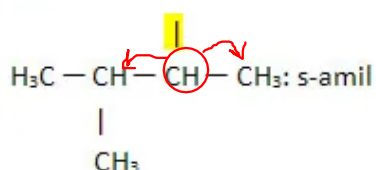
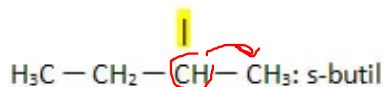
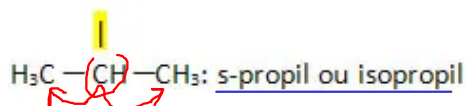


Nomenclatura quanto à valência livre

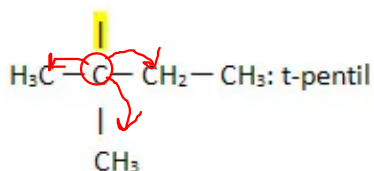
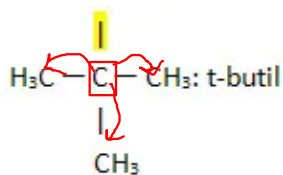
Iso: valência livre está localizada no carbono primário de uma cadeia ramificada



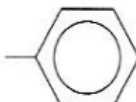
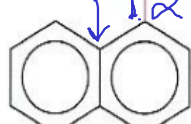
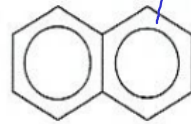
Sec- ou s-: valência livre no carbono secundário



Terc- ou t-: esse prefixo é utilizado para indicar que a valência livre está situada no carbono terciário

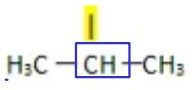
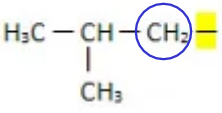
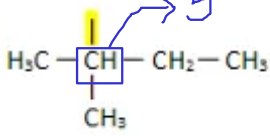
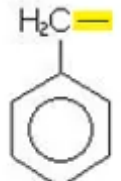


1 carbono	metil	$-\text{CH}_3$		
2 carbonos	etil	$-\text{CH}_2-\text{CH}_3$		
3 carbonos	propil (anteriormente: <i>n</i> -propil)	$-\text{CH}_2-\text{CH}_2-\text{CH}_3$	isopropil	$\begin{array}{c} -\text{CH}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$
4 carbonos	butil (anteriormente: <i>n</i> -butil)	$-\text{CH}_2(\text{CH}_2)_2\text{CH}_3$	isobutil	$\begin{array}{c} -\text{CH}_2-\text{CH}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$
	<i>s</i> -butil (<i>sec</i> -butil)	$\begin{array}{c} -\text{CH}-\text{CH}_2-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$	<i>t</i> -butil (<i>terc</i> -butil)	$\begin{array}{c} \text{CH}_3 \\ \\ -\text{C}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$
5 carbonos	pentil	$-\text{CH}_2(\text{CH}_2)_3\text{CH}_3$	isopentil	$-\text{CH}_2-\text{CH}_2-\text{CH}(\text{CH}_3)=\text{CH}_2$
	neopentil	$\begin{array}{c} \text{CH}_3 \\ \\ -\text{CH}_2-\text{C}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$	<i>t</i> -pentil (<i>terc</i> -pentil)	$\begin{array}{c} \text{CH}_3 \\ \\ -\text{C}-\text{CH}_2-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$

vinil ou etenil	$-\text{CH}=\text{CH}_2$	isopropenil	$\begin{array}{c} -\text{C}=\text{CH}_2 \\ \\ \text{CH}_3 \end{array}$
propen-1-il	$-\text{CH}=\text{CH}-\text{CH}_3$	alil ou propen-2-il	$-\text{CH}_2-\text{CH}=\text{CH}_2$
fenil		naft-1-il	
benzil	$-\text{CH}_2-$ 	naft-2-il	

Exercícios

Indique a alternativa que apresenta respectivamente os nomes corretos das seguintes ramificações:

- I.  isopropil = s-propil
- II.  isobutil 
- III.  fenil
- IV.  t-pentil
- V.  benzil

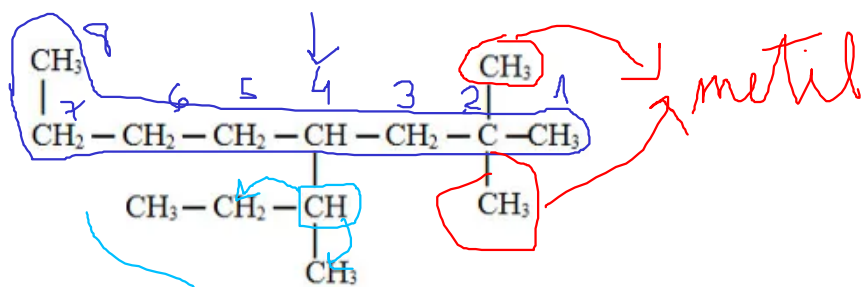
a) sec-butil, isopentil, benzil, terc-butil e vinil.

b) iso-propil, sec-butil, fenil, terc-pentil e benzil.

c) iso-propil, sec-butil, benzil, terc-pentil e fenil.

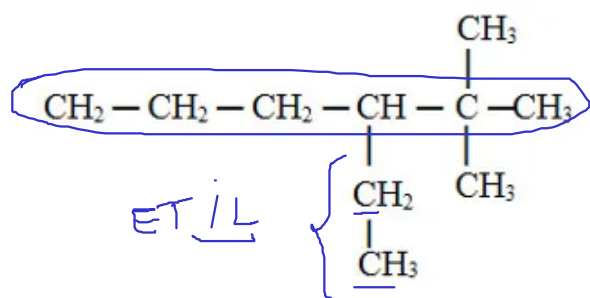
d) sec-propil, isobutil, benzil, terc-pentil e fenil.

e) sec-propil, isobutil, fenil, terc-pentil e benzil.

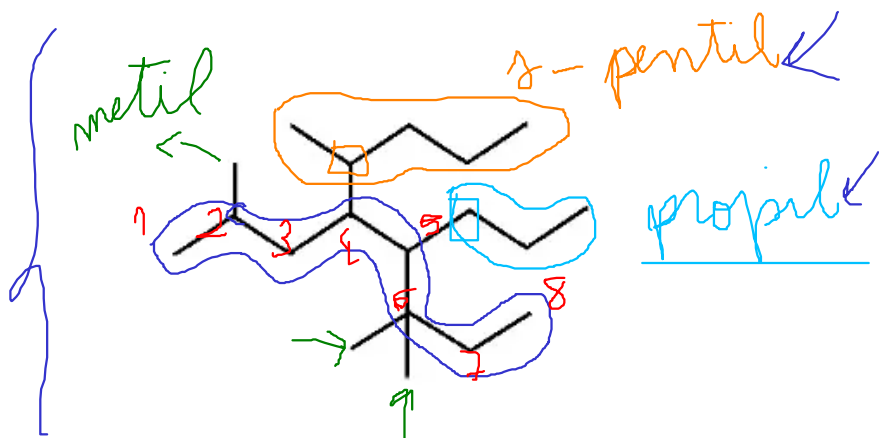


4-*s*-butil-2,2-dimetil-octano

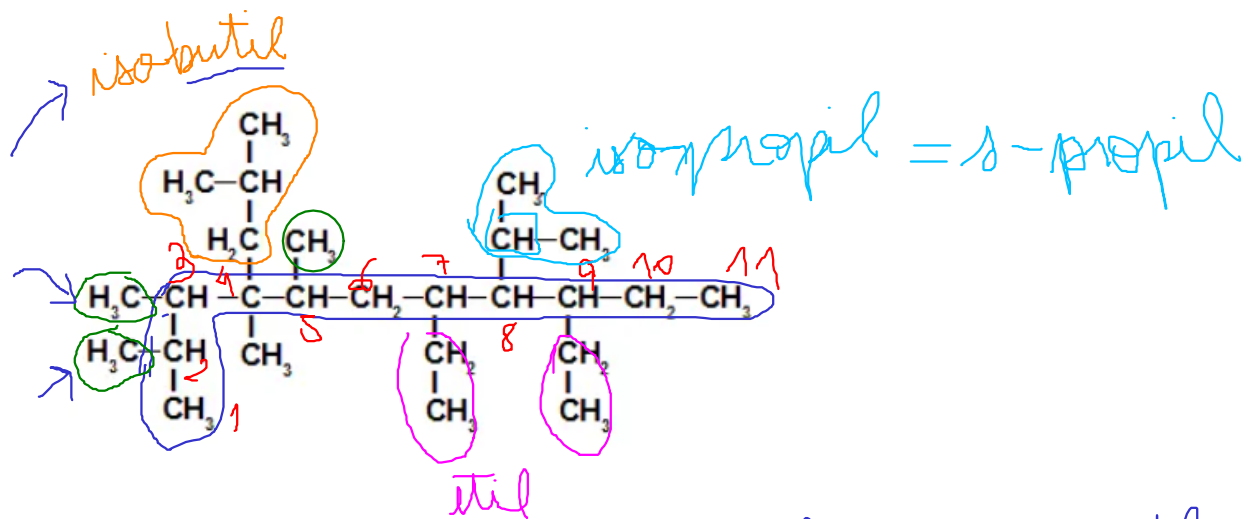
IUPAC



3-ethyl-2,2-dimethyl
 hexane

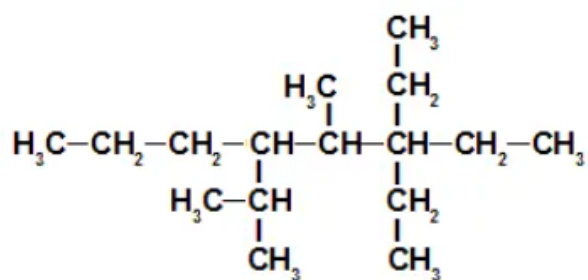


2,6,6-trimetil-4-5-pentil-5-propil-otano



4-isobutyl-7,9-diethyl-8-isopropyl-2,3,5-trimethylundecane

(Cesgranrio) Assinale a única alternativa correta com relação ao composto que apresenta a estrutura a seguir:



- a) é um alqueno.
- b) apresenta 1 radical n-propila ligado ao carbono 4.
- c) apresenta 2 radicais propila.
- d) apresenta 3 radicais etila.
- e) apresenta 2 radicais etila.